

Day 1

Define:-

A social science a govt. made calculation (managed allocated their resources) unlimited wants corresponding limited resources.

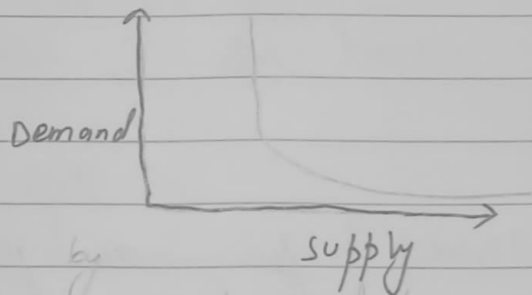
→ There is no economics if the resources are infinite.

free goods → are those goods which are available without any effort or money

non free goods → are those goods which are available only with efforts or money.

Demand & Supply → If demand is high and supply is low the price will be high and vice versa.

→ Both are inversely proportional to each other and affect the price.



→ We can evaluate a project that is it will be successful in future can be done by different analysis. which we study later.

Factors of Production:-

There are three factors

- 1) Land → earth space & natural resources
- 2) Labour → human inputs
- 3) Capital stock → man made item required for production

Day 2

Interest Rate & Rate of return :-

• Time value of money:

The value of money changes with time.

⇒ Interest Rate is the % of interest amount
It will be in %age and interest will be in dollars

I → Interest (\$, R)

i → interest-Rate (%)

Example :

① 10,000 \$ 1 year later 10,700 \$
Find interest and interest rate

$$\text{interest} = 10700 - 10000$$

$$= 700 \$$$

$$\text{interest rate} = \frac{700}{10,000} \times 100\%$$

$$= 7\%$$

② 20,000 \$ for one year interest rate 9%

$$\text{Interest} = 1800 \quad (\text{using interest rate formula})$$

$$\text{total amount} = 20000 + 1800$$

$$= 21800 \$$$

③ interest rate 5% return 1000 \$

amount owned = 1000 \$

interest rate = 5%

1000 = Principal + interest

1000 = Principal + Principal (interest Rate)

$$1000 = P(1 + 0.05)$$

$$\frac{1000}{1.05} = P$$

$$P = 952.38 \$$$

Terminology & Symbol

P → Value of money at Present or time (0)

F → Value of money at some future time

A → series of consecutive, equal, end of period, amount of money

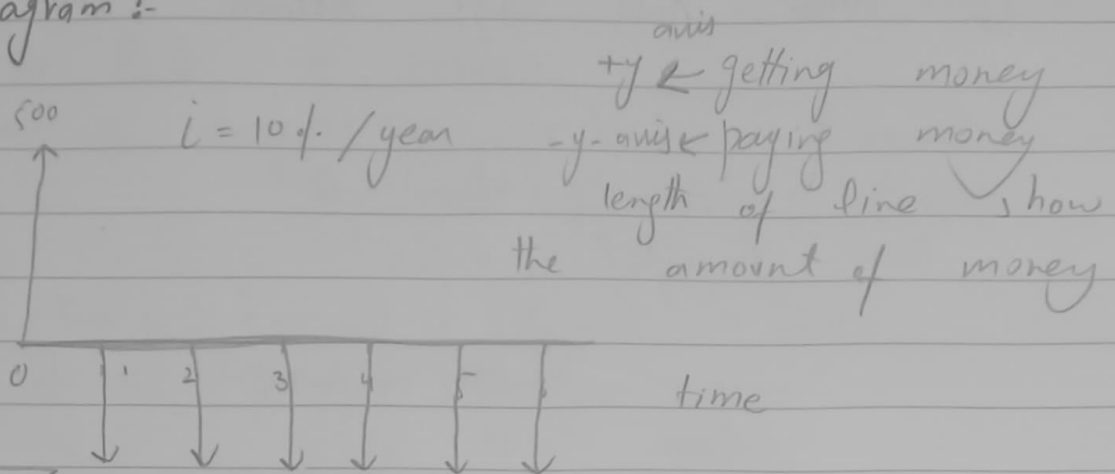
i → interest rate per time period

n → number of interest periods

t → time stated in period

∴ Present value of money is always larger than the Present

Cash flow diagram:-



If there is an interest rate then every amount have different value. These values can't be added cause all have different value but we can add and subtract the value at same time / period.

Example:-

